

Supplementary Materials

Leakage-controlled and deployment-aware input-resolution selection for dense broiler detection in precision poultry monitoring

Journal: Precision Agriculture

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This supplementary document contains additional leakage-audit, diagnostic, counting-calibration, repeatability, cross-detector robustness, external directional protocol-transfer, deployment-benchmarking, multi-objective resolution-selection, weight-sensitivity, and qualitative materials supporting the manuscript. Values are reported for the fixed leakage-controlled PIO-GRDB-MD5-7_1_2 split unless otherwise stated.

Supplementary and online resource map

Resource	Contents
Online Resource 1	Cross-split leakage audit, near-duplicate audit, component-level split diagnostics, and dataset split statistics.
Online Resource 2	Maximum-detection sensitivity and strict diagnostic AP analysis, including strict global-first scale and density diagnostic AP and TP/FN/FP summaries.
Online Resource 3	Counting calibration and confidence-threshold sensitivity, including fixed-threshold counting, test-set diagnostic sweeps, and validation confidence-MAE curves.
Online Resource 4	Seed repeatability, cross-detector robustness, locked-weight deployment benchmarking, multi-objective resolution selection, weight-sensitivity analysis, external VisDrone directional protocol-transfer demonstration source data, and qualitative examples, including low-/medium-density no-gain visual cases.
Online Resource 5	Bootstrap uncertainty and delta-difference analysis for resolution-dependent mAP50-95 differences.

Supplementary Tables

Table S1. Cross-split leakage audit.

Check item	Leakage key	Result	Interpretation
Source path	Absolute source image path	PASS	No cross-split leakage
Filename	Original basename	PASS	No cross-split leakage
Exact duplicate	Image-content MD5	PASS	No cross-split leakage
Source group	Inferred source group	PASS	No cross-split leakage

Note: All four leakage checks passed. The component warning reflects large source-related sequences rather than confirmed cross-split leakage.

Table S2. Near-duplicate audit using pHash/dHash and SSIM.

Method	Candidate rule	SSIM rule	SSIM size	Total	Train	Val	Test	Cross-split pairs	Hash candidates	Confirmed	Result
pHash/dHash candidate search + SSIM verification	pHash <= 6 OR dHash <= 6	SSIM >= 0.95	256x256 grayscale	1,487	1,027	153	307	519,391	6,517	0	PASS

Note: Candidate pairs were first screened by pHash/dHash and then verified by SSIM.

Table S3. Source-component-level split diagnostics for leakage-controlled splitting.

Split	Components	Images	Instances	Largest component images	Largest component instances
Train	361	1,027	225,771	352	150,312
Validation	2	153	36,905	130	35,690
Test	20	307	64,612	151	29,961

Note: Source components were inferred from source-path and filename-pattern metadata and kept within a single split to reduce source-related leakage risk. The largest-component columns document source-group concentration and split imbalance transparently.

Table S4. Full max_det sensitivity results.

Resolution	max_det	Precision	Recall	mAP50	mAP50-95
800	300	0.9377	0.8683	0.8903	0.6733
800	1000	0.9301	0.9283	0.9643	0.7199
800	1500	0.9301	0.9283	0.9643	0.7199
960	300	0.9390	0.8758	0.8933	0.6925
960	1000	0.9314	0.9385	0.9739	0.7420
960	1500	0.9314	0.9385	0.9739	0.7420
1280	300	0.9419	0.8790	0.8929	0.7025
1280	1000	0.9350	0.9438	0.9741	0.7543
1280	1500	0.9350	0.9438	0.9741	0.7545

Note: max_det=300 underestimated performance in high-density scenes; max_det=1000 and 1500 were nearly identical. At 1280 pixels, mAP50 saturated at max_det=1000 and showed no further change at 1500.

Table S5. Strict global-first diagnostic mAP50-95 for scale-defined subgroups.

Scale group	Images	Instances	800	960	1280
tiny	279	16,153	0.1249	0.1366	0.1414
small	300	16,153	0.1613	0.1678	0.1738
medium	289	16,153	0.2499	0.2693	0.2521
large	278	16,153	0.3972	0.3571	0.4474

Note: This is a strict global-first diagnostic mAP50-95. It is computed after global prediction ranking and global matching, followed by attribution of matched true positives to scale-defined subgroups. It is contribution-oriented and should not be numerically compared with standard global AP or ordinary scale AP.

Table S6. Strict global-first diagnostic mAP50-95 for density-defined subgroups, with strict-pipeline global mAP50-95 as a scale reference.

Resolution	Strict-pipeline global mAP50-95	Low	Medium	High	Ultra-high
800	0.7183	0.0860	0.1476	0.2299	0.2940
960	0.7389	0.0762	0.1344	0.2622	0.3170
1280	0.7539	0.0750	0.1460	0.2593	0.3170

Note: The strict-pipeline global mAP50-95 column is included only as a numerical scale reference for this diagnostic pipeline. These values are computed within the strict diagnostic pipeline and may differ marginally from the standard test-set values in Table 4 of the main manuscript because of implementation-level ranking aggregation; the authoritative standard global results remain those in the main manuscript Table 4. Strict global-first diagnostic mAP50-95 is computed after global prediction ranking and global one-to-one matching over the full test set, followed by attribution of matched true positives to density-defined subgroups. Because the prediction ranking is not re-optimized within each subgroup, these values are contribution-oriented diagnostic scores and are not numerically comparable with ordinary subgroup AP values reported in the main text.

Table S7. TP/FN/FP by density group.

Density	Resolution	TP	FN	FP	Recall	Precision
low	800	6,907	189	1,701	0.9734	0.8024
low	960	6,890	206	1,629	0.9710	0.8088
low	1280	6,915	181	1,472	0.9745	0.8245
medium	800	11,896	404	1,766	0.9672	0.8707
medium	960	11,821	479	1,664	0.9611	0.8766
medium	1280	11,839	461	1,451	0.9625	0.8908
high	800	17,303	621	2,679	0.9654	0.8659
high	960	17,356	568	2,291	0.9683	0.8834
high	1280	17,385	539	2,068	0.9699	0.8937
ultra-high	800	26,399	893	3,944	0.9673	0.8700
ultra-high	960	26,612	680	3,579	0.9751	0.8815
ultra-high	1280	26,605	687	3,079	0.9748	0.8963

Note: Computed at confidence threshold=0.25 and IoU threshold=0.50. Recall is computed as $TP/(TP+FN)$, and precision is computed as $TP/(TP+FP)$.

Table S8. Fixed-threshold counting at confidence threshold 0.25.

Resolution	Conf	Images	GT total	Pred total	MAE	RMSE	MAPE (%)	Bias	Rel. total error (%)
800	0.25	307	64,612	72,491	26.07	34.43	13.72	+25.66	+12.19
960	0.25	307	64,612	71,759	24.56	32.46	12.94	+23.28	+11.06
1280	0.25	307	64,612	70,822	21.23	30.07	11.41	+20.23	+9.61

Note: This fixed threshold was used to align counting with the TP/FN/FP diagnostic and qualitative visualization protocol.

Table S9. Density-specific fixed-threshold counting at confidence threshold 0.25.

Resolution	Density	Images	GT total	Pred total	MAE	RMSE	MAPE (%)	Bias
800	low	78	7,096	8,596	19.49	31.55	20.89	+19.23
800	medium	76	12,300	13,649	18.43	22.98	11.40	+17.75
800	high	76	17,924	19,951	27.38	32.31	11.58	+26.67
800	ultra-high	77	27,292	30,295	39.00	46.54	10.88	+39.00
960	low	78	7,096	8,503	18.91	30.01	20.15	+18.04
960	medium	76	12,300	13,462	18.08	22.26	11.21	+15.29
960	high	76	17,924	19,635	23.99	27.66	10.16	+22.51
960	ultra-high	77	27,292	30,159	37.23	45.21	10.09	+37.23
1280	low	78	7,096	8,387	16.73	28.33	18.41	+16.55
1280	medium	76	12,300	13,296	15.53	19.64	9.68	+13.11
1280	high	76	17,924	19,454	21.05	26.06	8.93	+20.13
1280	ultra-high	77	27,292	29,685	31.60	41.73	8.47	+31.08

Note: Density groups are defined by the number of GT instances per full test image.

Table S10. Test-set counting confidence sweep, diagnostic only.

Resolution	Best conf	Images	GT total	Pred total	MAE	RMSE	MAPE (%)	Bias	Rel. total error (%)
800	0.45	307	64,612	64,735	12.30	18.31	7.33	+0.40	+0.19
960	0.40	307	64,612	66,096	13.31	19.64	8.17	+4.83	+2.30
1280	0.40	307	64,612	66,264	12.70	19.62	7.59	+5.38	+2.56

Note: Diagnostic only: these thresholds were selected by sweeping the test set for sensitivity analysis and are not the validation-calibrated thresholds used for the main counting results in Table 9 of the main manuscript.

Table S11. Seed-repeatability table location. The YOLO11n seed-repeatability table is reported in the main manuscript as Table 11 and is not duplicated here to avoid redundancy.

Note: See the main manuscript Table 11 for seed-level values and the early-stopping note for the 1280-pixel seed-2024 run.

Supplementary Figures

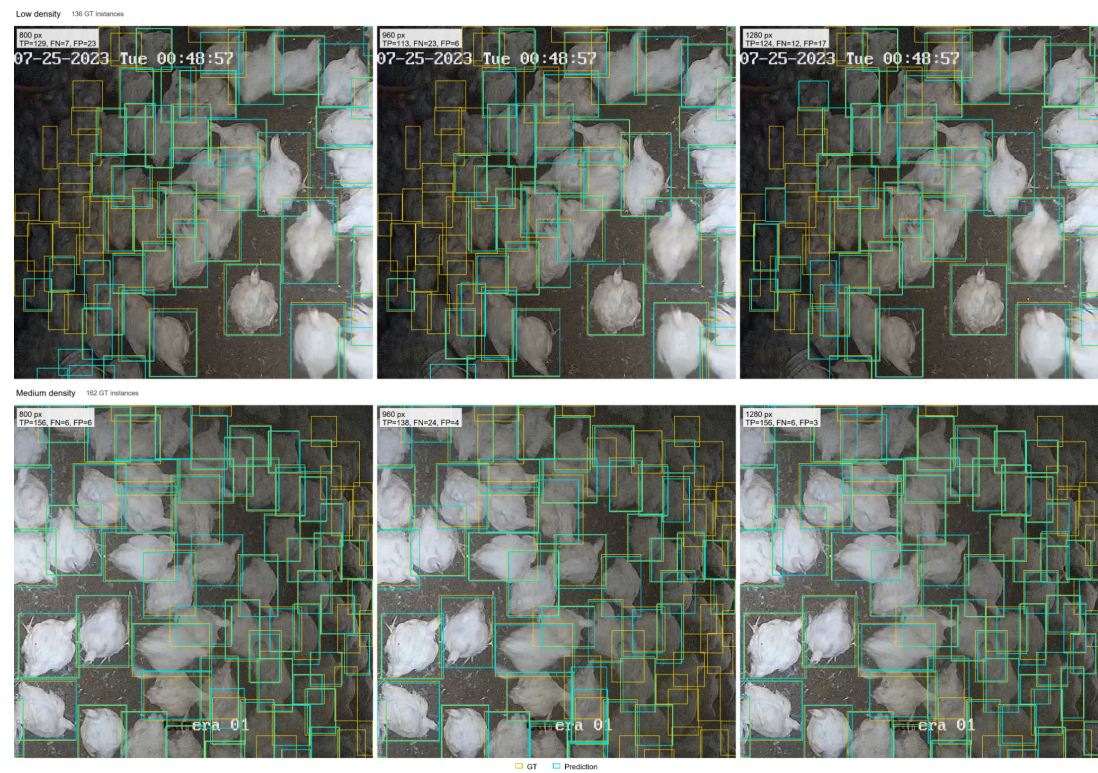


Figure S1. Qualitative examples from the low- and medium-density subsets where higher input resolution did not provide a clear advantage. Yellow boxes indicate ground-truth annotations, and cyan boxes indicate predictions. TP, FN, and FP were computed at confidence threshold = 0.25 and IoU threshold = 0.50, while only predictions with confidence ≥ 0.50 are displayed for readability. Density groups were defined based on the full-image instance count, although only local crops are shown for visual clarity.

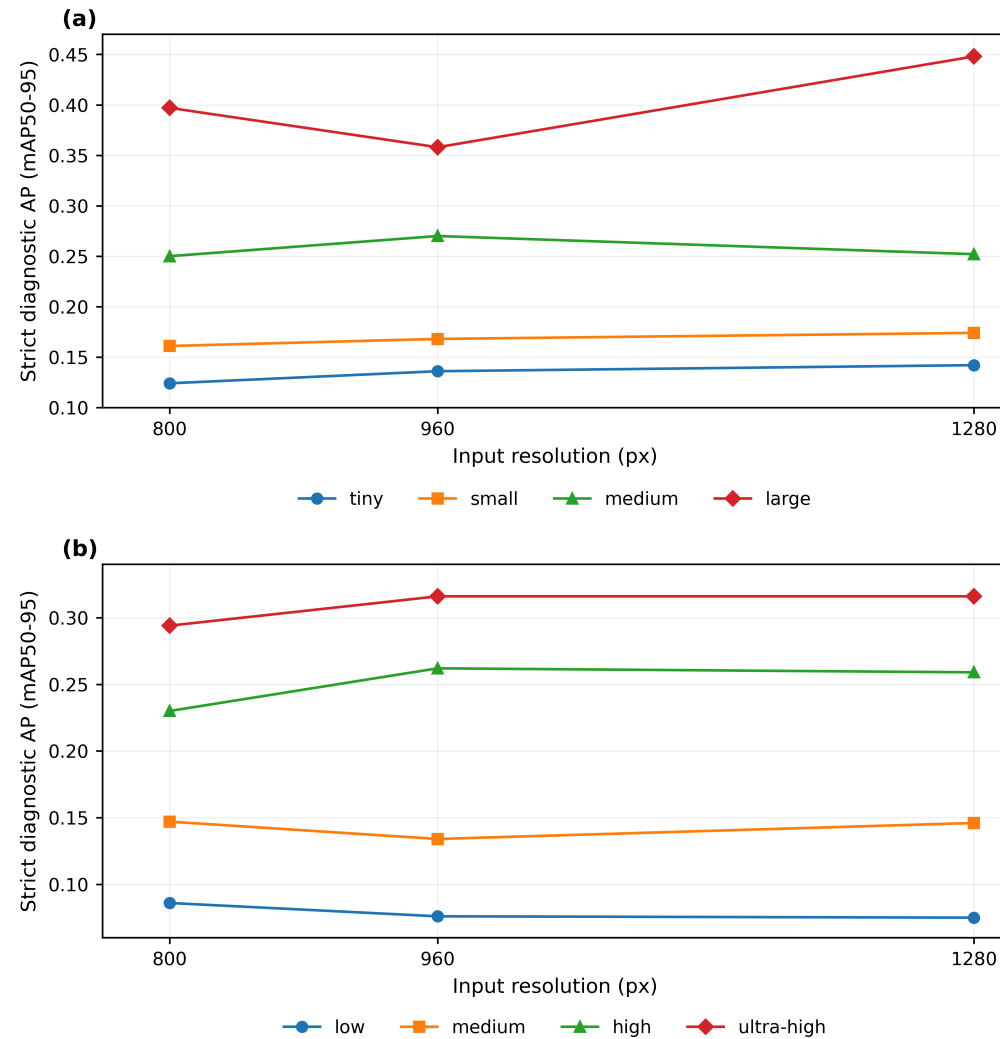


Figure S2. Strict global-first diagnostic AP by (a) target scale group and (b) image density group. Strict diagnostic AP preserves the global ranked prediction list and global matching results before attributing matched true positives to predefined groups. Therefore, it measures subgroup contributions under the full-test-set operating order and is intended for within-protocol diagnostic comparison only. The values are not directly comparable with ordinary subgroup AP values in Figs. 3 and 4 or with standard global AP.

Additional supporting analyses

The following analyses provide additional support for repeatability, cross-detector robustness, deployment benchmarking, multi-objective resolution selection, weight-sensitivity analysis, external directional protocol transfer, marginal-gain reduction, and validation-based counting-threshold calibration. Figure S3 was generated from validation prediction CSV files and validation labels.

Table S12a. Paired bootstrap mAP50-95 delta and delta-difference analysis for marginal gain reduction.

Quantity	Mean	Median	95% CI low	95% CI high	SD	P(Delta < 0)	P(Delta > 0)	n
Delta(960-800)	0.019464	0.019531	0.016652	0.022146	0.001454	0.000	1.000	1000
Delta(1280-960)	0.013562	0.013570	0.011330	0.015651	0.001113	0.000	1.000	1000
Delta(1280-960) - Delta(960-800)	-0.005902	-0.005981	-0.010364	-0.001518	0.002262	0.998	0.002	1000

Note: Delta-difference is defined as Delta(1280-960) - Delta(960-800). A negative value indicates that the gain from 960 to 1280 is smaller than the gain from 800 to 960. For mAP50-95, both resolution increments had 95% confidence intervals above zero, and the delta-difference had a 95% confidence interval below zero, supporting the marginal-gain reduction interpretation.

Table S12b. Paired bootstrap AP50 delta and delta-difference analysis.

Quantity	Mean	Median	95% CI low	95% CI high	SD	P(Delta < 0)	P(Delta > 0)	n
Delta(960-800)	0.005231	0.004385	0.002658	0.009075	0.002050	0.000	1.000	1000
Delta(1280-960)	0.000865	0.000879	-0.000187	0.001886	0.000528	0.054	0.946	1000
Delta(1280-960) - Delta(960-800)	-0.004366	-0.003790	-0.008622	-0.001087	0.002167	1.000	0.000	1000

Note: AP50 was close to saturation. The 960-to-1280 AP50 gain was small and its 95% confidence interval included zero, indicating that the additional benefit of 1280 pixels was more evident for stricter localization performance (mAP50-95) than for coarse IoU=0.50 detection.

Table S13. Cross-detector robustness check using YOLOv8n under the same leakage-controlled resolution protocol.

Resolution	Training (h)	Val P	Val R	Val mAP50	Val mAP50-95	Test P	Test R	Test mAP50	Test mAP50-95
800	4.908	0.9496	0.9509	0.9760	0.7465	0.9328	0.9356	0.9660	0.7272
960	6.641	0.9516	0.9570	0.9846	0.7710	0.9361	0.9416	0.9733	0.7443
1280	13.984	0.9543	0.9580	0.9853	0.7862	0.9380	0.9431	0.9740	0.7573

Note: YOLOv8n was used only as a cross-detector robustness check under the same leakage-controlled split and resolution protocol. It was not used for detector ranking, seed-repeatability analysis, or the main resolution-selection conclusion. Latency columns were excluded from this table because the initial single-run latency summary contained a timing anomaly; deployment conclusions were based on the repeated locked-weight YOLO11n benchmark.

Table S14. YOLO11n locked-weight deployment benchmark using the 30-repeat recheck protocol.

Resolution (px)	Params	GFLOPs	Val mAP50-95	Preprocess (ms/img)	Inference (ms/img)	Postprocess (ms/img)	Total latency (ms/img)	Total IQR (ms/img)	Peak allocated (MiB)	Peak reserved (MiB)
800	2,590,035	10.063	0.7387	1.083	3.943	1.303	6.324	0.331	60.7	86.0
960	2,590,035	14.491	0.7680	1.496	5.365	1.638	8.520	0.192	75.1	102.0
1280	2,590,035	25.762	0.7824	1.138	6.499	1.523	9.146	0.669	97.4	142.0

Note: The benchmark was conducted on the validation split with batch=2, workers=0, confidence threshold=0.001, NMS IoU=0.70, max_det=1000, and FP16 GPU inference. Five warm-up repeats were discarded, and medians and IQRs were computed from thirty formal repeats. Total latency is the median end-to-end time per image across formal repeats and may not equal the sum of individual stage medians because each stage median is computed independently. Peak allocated refers to the peak CUDA memory actively allocated by tensors, whereas peak reserved refers to the peak CUDA memory reserved by the PyTorch caching allocator. This 30-repeat recheck replaces the earlier five-repeat latency summary.

Table S15. Multi-objective resolution-selection analysis using the 30-repeat deployment recheck.

Resolution (px)	Test mAP50-95	Training time (h)	GFLOPs	Total latency (ms/img)	Peak reserved (MiB)	Default score	Rank	Interpretation
800	0.7199	4.48	10.063	6.324	86.0	0.500	2	Lowest-cost baseline with the lowest accuracy
960	0.7420	7.29	14.491	8.520	102.0	0.656	1	Balanced candidate under the default weighted objective
1280	0.7543	22.54	25.762	9.146	142.0	0.500	2	Accuracy-oriented setting with the highest GFLOPs and memory pressure

Note: The default score was computed as $\text{Score} = 0.50 \cdot \text{Accuracy_norm} + 0.20 \cdot \text{TrainingEfficiency_norm} + 0.10 \cdot \text{GFLOPsEfficiency_norm} + 0.10 \cdot \text{LatencyEfficiency_norm} + 0.10 \cdot \text{MemoryEfficiency_norm}$. Accuracy was normalized as a benefit metric, whereas training time, GFLOPs, latency, and memory were normalized as cost-efficiency metrics. The score is not intended as a universal utility function; it provides one explicit task-objective-dependent example. The 800- and 1280-pixel settings had equal default scores and were therefore assigned the same rank.

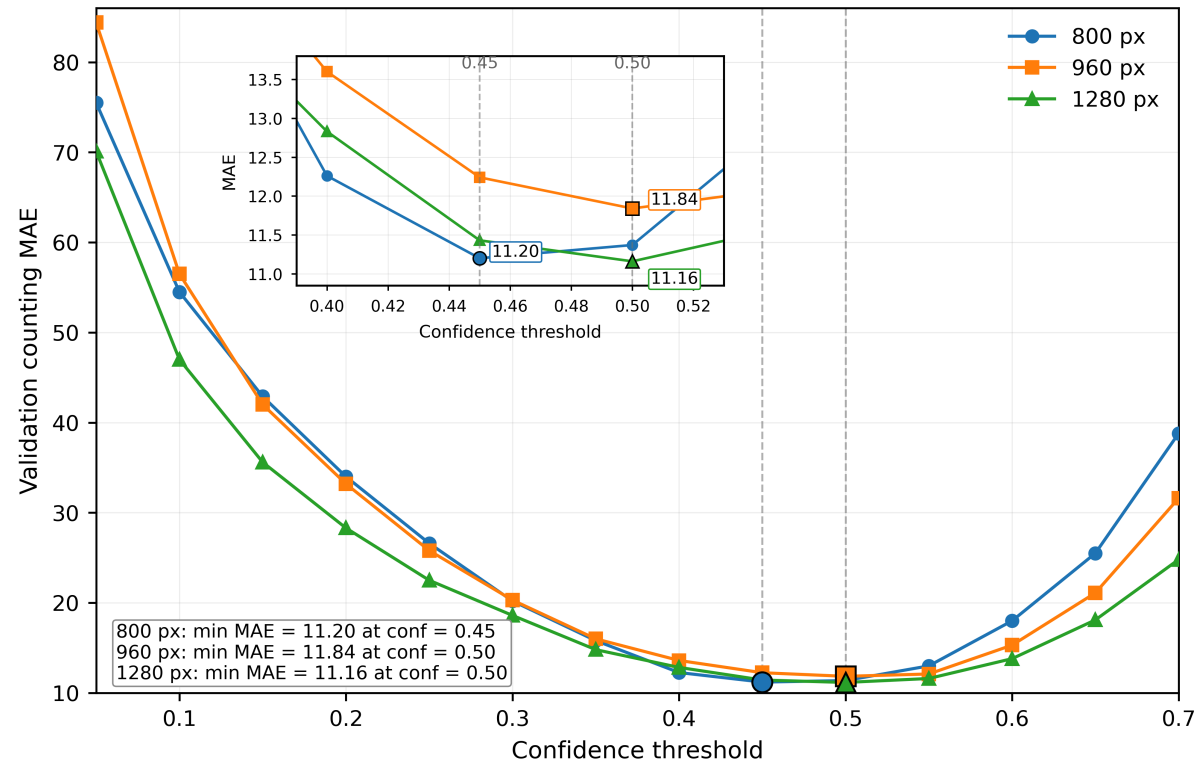


Figure S3. Validation confidence sweep for counting-threshold calibration. The main panel preserves the continuous full-range curve, while the inset magnifies the selected-threshold region. Validation-calibrated thresholds were selected from the validation set and then applied to the held-out test set.

Table S16. Sensitivity of the multi-objective resolution selection to the accuracy weight.

Accuracy weight α	Selected resolution	Interpretation
$0.000 \leq \alpha < 0.340$	800	Cost-prioritized objective; lowest-cost baseline selected
$0.340 \leq \alpha \leq 0.652$	960	Balanced objective; intermediate resolution selected
$0.652 < \alpha \leq 1.000$	1280	Accuracy-prioritized objective; highest-accuracy setting selected
sample $\alpha = 0.00$	800	score800=1.000; score960=0.669; score1280=0.000
sample $\alpha = 0.25$	800	score800=0.750; score960=0.662; score1280=0.250
sample $\alpha = 0.50$	960	score800=0.500; score960=0.656; score1280=0.500
sample $\alpha = 0.75$	1280	score800=0.250; score960=0.649; score1280=0.750
sample $\alpha = 1.00$	1280	score800=0.000; score960=0.642; score1280=1.000

Note: The sensitivity score was defined as $\text{Score}_\alpha = \alpha \text{ Accuracy_norm} + (1 - \alpha) \text{ CostEfficiency_mix}$, where the internal cost-efficiency mixture kept the ratio Training:GFLOPs:Latency:Memory = 0.4:0.2:0.2:0.2. This analysis shows that the 960-pixel setting is not universally optimal; it is selected only within a moderate accuracy-prioritized objective range. Cost-prioritized settings select 800 pixels, whereas strongly accuracy-prioritized settings select 1280 pixels.

Table S17. External cross-domain resolution demonstration on VisDrone2019-DET using YOLO11n.

Resolution (px)	Training time (h)	Val P	Val R	Val mAP50	Val mAP50-95	Test-dev P	Test-dev R	Test-dev mAP50	Test-dev mAP50-95
800	8.70	0.4713	0.3885	0.3763	0.2180	0.4216	0.3437	0.3146	0.1783
960	10.28	0.5223	0.4122	0.4210	0.2488	0.4550	0.3755	0.3505	0.2007
1280	14.33	0.5679	0.4596	0.4797	0.2885	0.4981	0.4102	0.3977	0.2300

Note: This experiment was conducted as an external cross-domain protocol demonstration rather than as a VisDrone state-of-the-art benchmark. YOLO11n was trained on VisDrone2019-DET using seed=42, batch=2, workers=0, SGD, and 100 epochs for each input resolution. The official train, validation, and test-dev splits were used. The absolute AP values are substantially lower than those obtained on PIO because VisDrone2019-DET is a multi-class aerial dense small-object benchmark with smaller objects, heavier occlusion, larger viewpoint variation, and more diverse backgrounds. Nevertheless, both validation and test-dev results showed a directionally consistent resolution trend, with mAP50-95 increasing from 800 to 960 and 1280 pixels. An auxiliary duplicate audit was performed on the official VisDrone2019-DET train/validation/test-dev splits. The audit found zero cross-split basename-overlap pairs, zero cross-split image-MD5 duplicate pairs, and zero cross-split annotation-MD5 duplicate pairs. Six cross-split dHash near-duplicate candidates were identified at dHash Hamming distance ≤ 6 , but none reached the SSIM confirmation threshold of $\text{SSIM} \geq 0.95$. Therefore, no SSIM-confirmed cross-split near-duplicate leakage was detected. This audit was used only to support the external demonstration; the VisDrone splits were not reconstructed by the source-group splitting protocol used for the primary PIO case study.

Additional revision analyses for counting calibration

Table S18. Source-component sensitivity of validation-calibrated counting thresholds.

Resolution	Calibration subset	Calibration images	Selected confidence	Test MAE	Test RMSE	Test bias	Rel. total error (%)
800	Full validation	153	0.45	12.30	18.31	+0.40	+0.19
800	Component 1 only	130	0.45	12.30	18.31	+0.40	+0.19
800	Component 10 only	23	0.40	12.44	19.57	+6.34	+3.01
960	Full validation	153	0.50	15.72	22.26	-7.42	-3.53
960	Component 1 only	130	0.50	15.72	22.26	-7.42	-3.53
960	Component 10 only	23	0.35	15.22	22.36	+10.70	+5.09
1280	Full validation	153	0.50	14.01	20.49	-4.55	-2.16
1280	Component 1 only	130	0.50	14.01	20.49	-4.55	-2.16
1280	Component 10 only	23	0.45	12.89	19.23	+0.63	+0.30

Note: For each resolution, thresholds were selected either from the full validation set or from one validation source component only, then fixed and applied to the unchanged held-out test set. The validation split contained two components: component 1 with 130 images and component 10 with 23 images. Full-validation and dominant-component calibration produced identical test results, whereas the smaller component changed selected thresholds, indicating partial sensitivity to validation-source composition.

Table S19. Counting-threshold robustness under fixed, validation-calibrated, and diagnostic test-set thresholds.

Resolution	Protocol	Selection split	Selected confidence	Test MAE	Test RMSE	Test MAPE (%)	Test bias	Rel. total error (%)
800	Fixed conf. 0.25	None (fixed)	0.25	26.07	34.43	13.72	+25.66	+12.19
800	Validation calibrated	Validation	0.45	12.30	18.31	7.33	+0.40	+0.19
800	Test diagnostic best	Test	0.45	12.30	18.31	7.33	+0.40	+0.19
960	Fixed conf. 0.25	None (fixed)	0.25	24.56	32.46	12.94	+23.28	+11.06
960	Validation calibrated	Validation	0.50	15.72	22.26	8.79	-7.42	-3.53
960	Test diagnostic best	Test	0.40	13.31	19.64	8.17	+4.83	+2.30
1280	Fixed conf. 0.25	None (fixed)	0.25	21.23	30.07	11.41	+20.23	+9.61
1280	Validation calibrated	Validation	0.50	14.01	20.49	7.98	-4.55	-2.16
1280	Test diagnostic best	Test	0.40	12.70	19.62	7.59	+5.38	+2.56

Note: The fixed-threshold protocol used confidence = 0.25 and was included for diagnostic consistency with the TP/FN/FP and visualization analyses. The validation-calibrated protocol selected the threshold on the validation set and applied it once to the held-out test set; this is the formal counting protocol. The test-set best threshold was selected directly on the test set and is reported only as a diagnostic sensitivity analysis. It was not used for model selection or for the main counting conclusion.